La Scuola d'Italia / Huson / Physics Nickname:
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2.4 Classwork: Constant Acceleration Practice

Reference (provided). These hold when the acceleration a is constant.

$$v = v_0 + at \qquad v^2 = v_0^2 + 2a(x - x_0)$$

$$x = x_0 + v_0 t + \tfrac{1}{2}at^2 \qquad \bar{v} = \frac{v + v_0}{2}$$

For each problem, list what you know (v_0, v, a, t, $x - x_0$) and what you want, then choose the equation that connects them. Show the units.

1. The Vernier track from 2.2 is set at a gentler slope. The cart is released from rest and speeds up at a constant $a = 0.200$ m/s^2.

   (a) The velocity grows by 0.200 m/s every second. Fill in the table.

t (s)	0	1.00	2.00	3.00	4.00
v (m/s)					

   (b) How far has the cart rolled at $t = 3.00$ s?

2. The elevators in Taipei 101 were the fastest in the world from 2004 to 2016, with a top speed of 16.8 m/s. Suppose an elevator starts from rest and speeds up at a constant 1.20 m/s^2.

   (a) How long does it take to reach top speed?

   (b) What is its average velocity while it speeds up?

   (c) How far does it rise while speeding up?

3. A driver at 30 mi/h (13.4 m/s) sees a hazard ahead. The UK Highway Code allows 9 m of *thinking distance*, covered before the foot reaches the brake, and then 14 m of braking distance.

   (a) While the driver reacts, the car keeps going at 13.4 m/s, so $a = 0$. How long does the driver take to react?

   (b) In 2.3 you found that the car brakes at $a = -6.4$ m/s^2. How long does it take to stop?

   (c) What is the total time, and the total distance, from seeing the hazard to stopping?

4. The Frecciarossa 1000 runs between Rome and Milan at 300 km/h (83.3 m/s). Approaching Milano Centrale, suppose it slows down at a constant 0.500 m/s^2, so $a = -0.500$ m/s^2.

   (a) How long does the train take to stop?

   (b) How far from the station must the driver start braking? Give your answer in km.